Pavithra Raghavan



GitHub | Google Scholar | Website | Email
M.Sc. Computer Science, University of Bonn

Summary

M.Sc. Computer Science student at the University of Bonn specializing in computer vision, generative AI, and intelligent systems. Experienced in vision-language models, reinforcement learning, and flow-based generative modelling for 3D human pose estimation and motion forecasting. Research interests include human motion modelling, 3D scene understanding, multimodal learning, and visual perception for autonomous and embodied systems, with a broader interest in applying machine learning across scientific domains.

Education

2024 – present M.Sc. Computer Science, Rhenish Friedrich Wilhelm University of Bonn, Germany
Focus: Computer Vision and Intelligent Systems .

2019 - 2023 Bachelor of Engineering, Rajalakshmi Institute of Technology - Anna University, India

Research Experience

Student Research Assistant *May 2025 – 2026*
cciBonn, University Hospital Bonn

- Worked on multimodal vision–language models for radiology report generation from multi-view chest X-rays, including LlamaRG and InternVL-based models.
- Developed and evaluated reinforcement-learning-based post-training pipelines, exploring SCST, PPO, and GRPO and different reward formulations for clinical report quality.

Tech: PyTorch • Hugging Face • Vision–Language Models • SCST • PPO • GRPO

Research Projects

Reports & Presentations

Generative 3D Human Pose Estimation and Forecasting from Video *May 2026 – Present*
Master's Thesis, Computer Vision Group, University of Bonn

- Investigating flow-matching-based generative models for monocular 3D human pose estimation, including input-informed flow initialization and multi-hypothesis prediction.
- Developing a video-to-3D pose estimation pipeline with the goal of integrating temporal motion forecasting and studying uncertainty and error propagation.
- Evaluating 3D pose estimation methods on Human3.6M using MPJPE and PA-MPJPE, with comparisons to PHD and FMPose3D baselines.

Tech: PyTorch • Flow Matching • Human3.6M • 3D Pose Estimation • Motion Forecasting

Instance Segmentation of Buildings from Satellite Imagery *Nov 2025 – Mar 2026*
Vision Lab Coursework

- Developed a Mask R-CNN pipeline for building extraction from high-resolution satellite imagery. Improved boundary accuracy using contour-based refinement with spectral and geometric priors.

Tech: PyTorch • Mask R-CNN • OpenCV • Remote Sensing

Multilingual Article Classification *Nov 2024 – Jan 2025*
Transformer-based NLP Project

- Built multilingual document classifiers using XLM-R and mBERT with attention mechanisms for cross-lingual news classification. Improved robustness to class imbalance and multilingual generalization.

Tech: XLM-R • mBERT • Transformers • PyTorch

Personalized Sequential POI Recommendation *Apr 2024 – Jul 2024*
Sequential Recommendation Project

- Designed an LSTM-based sequential recommendation model using spatial and temporal feature engineering. Achieved 58% venue prediction and 99% venue-type prediction accuracy.

Tech: PyTorch • LSTM • Sequential Recommendation • Geo-spatial Features

Deepfake Video Detection *Feb 2023 – May 2023*
Computer Vision Project

- Developed a frame-level deepfake detection pipeline using handcrafted visual features and an MLP classifier. Evaluated performance on the FaceForensics++ benchmark.

Tech: Python • OpenCV • MLP • FaceForensics++

Publications and Preprints

- T. Jayas, A. Rastogi, **P. Raghavan**, G. Brugnarra, K. Schlamp, M. Foltyn-Dumitru, and P. Vollmuth. *LlamaRG: A Multi-View Large Language Model for Radiology Report Generation.* *CVPR Findings*, 2026, pp. 5745–5754.
- S. R. S, B. Sriman, **P. Raghavan**, N. R. G, T. Sunitha, G. Thiyagarajan. *"Analysis on Semantic Categorization Technique of Analogous Label Using Deep Learning,"* In *Proc. International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, 2022.
  IEEE Xplore

Technical Profile

Programming	Python, C
Frameworks	PyTorch, Hugging Face Transformers, OpenCV, NumPy, Pandas
Methods	Flow Matching, Generative Modelling, Reinforcement Learning, Vision–Language Models, Transformers
Computer Vision	3D Human Pose Estimation, Human Motion Forecasting, Instance Segmentation, Medical Image Analysis
Tools	Git, Linux, CUDA, LaTeX
Research Interests	3D Computer Vision, Generative AI, Dynamic 3D Scene Understanding, Human Motion Modelling, Embodied Perception
Relevant Coursework	Computer Vision, Machine Learning, Reinforcement Learning, Geometric Deep Learning, Natural Language Processing, Humanoid Robotics